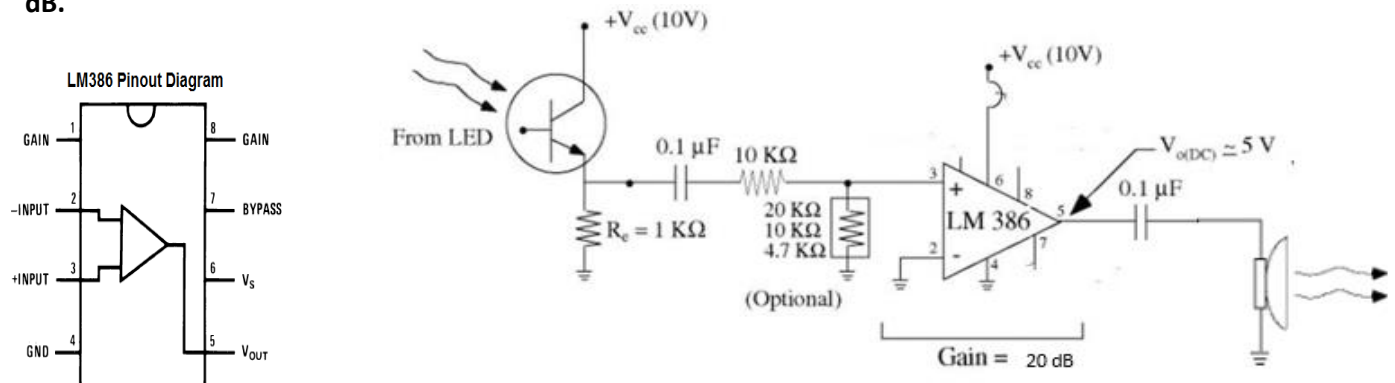
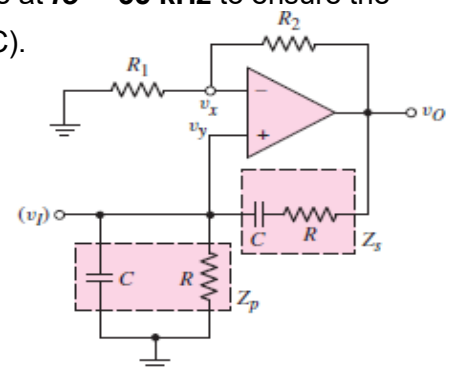


- Quartz-crystal oscillators are based on
(a) positive feedback (b) negative feedback (c) the piezoelectric effect (d) high gain
- The term *pole* in filter terminology refers to
(a) a single *RC* circuit (b) one complete active filter (c) a high-gain op-amp (d) the feedback circuit
- An oscillator differs from an amplifier because the oscillator
(a) has more gain (b) requires no input signal (c) requires no dc supply (d) always has the same output
- When a low-pass and a high-pass filter are cascaded to get a band-pass filter, **the critical frequency (f_c) of the low-pass filter must be**
(a) equal to the critical frequency of the high-pass filter (b) less than the critical frequency of the high-pass filter (c) greater than the critical frequency of the high-pass filter.
- The main feature of a crystal oscillator is
(a) economy (b) reliability (c) high frequency (d) stability
- An oscillator whose frequency is changed by a variable dc voltage is known as
(a) a crystal oscillator (b) a VCO (c) an Armstrong oscillator (d) a piezoelectric device

Q2. In the interest of time, the FM receiver is built for you. This will let you concentrate on measuring the performance of the receiver and on listening to FM music. The output of the phototransistor is first passed by a capacitor to block the DC and then amplified by the LM 386. **The problem is that the LM 386 has a minimum gain of 20. Develop the circuit in the figure below to provide the variable gain from 20 dB- 200 dB.**



Q3. Design the Wein-bridge oscillator in Figure below , to oscillate at **$f_o = 35 \text{ kHz}$** to ensure the startup of oscillation, you should have $(R_2/R_1 = 2)$ and **$\omega_o = (1/RC)$** .



ANSWER

- 1.(C)
2. (a)
3. (b)
- 4.(c)
5. (d)
- 6.(b)

Q2. Connect variable resistance a series with a capacitor 10uf pin (1 and 8).

The output signal from the volume control knob is given to the input of the non-inverting input pin of the audio amplifier IC (LM386) through a coupling capacitor of $0.01\ \mu\text{F}$ to remove any DC component present in the signal. LM386 is a low voltage audio amplifier IC whose maximum output power of is about 1 W which is enough to drive headphones or earphones of 1 W rating. The internal architecture of LM386 is as shown in Fig. 1. The inverting input pin of LM386 is connected to the ground terminal. A bypass capacitor can be connected to the bypass pin of LM386 if required. Bypass capacitor is used to remove any AC interference noise if present. The power supply used for this IC is the same 9 V battery. A capacitor of $470\ \text{pF}$ is placed across the non-inverting input terminal and ground to remove any radio frequency interferences and a capacitor of $0.1\ \mu\text{F}$ is placed across the V_{cc} pin and ground pin of LM386 to decouple any noise entering the IC. The internal minimum gain for the power amplifier IC is set to 20. The gain can be varied by placing a combination of capacitor and resistor in series across the gain pins of the IC. The maximum gain that can be achieved by LM386 is about 200 by placing only a capacitor of $10\ \mu\text{F}$ across the gain pins [9].

The internal minimum gain for the power amplifier IC is set to 20. The gain can be varied by placing a combination of capacitor and resistor in series across the gain pins of the IC. The maximum gain that can be achieved by LM386 is about 200 by placing only a capacitor of $10\ \mu\text{F}$ across the gain pins. The output amplified signal from the IC LM386 is passed through a parallel combination of Zobel filter. In this device, the Zobel filter is formed by connecting a capacitor of $0.047\ \mu\text{F}$ and resistor of $10\ \Omega$ in series and connecting the capacitor end to the output of the power amplifier IC and shorting the resistor end to ground. It is a parallel circuit of series resistor
